DTU Compute
Department of Applied Mathematics and Computer Science

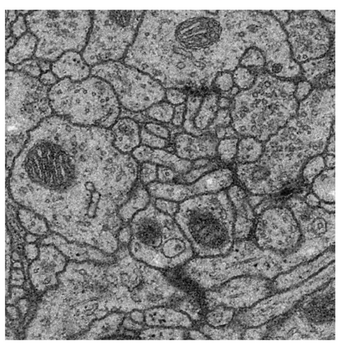
Segmentation of Membranes in Electron Microscopy Images Using Convolutional Neural Networks

Casper Brun Pedersen , **Mads Prip** & **Petr Boska Nylander**

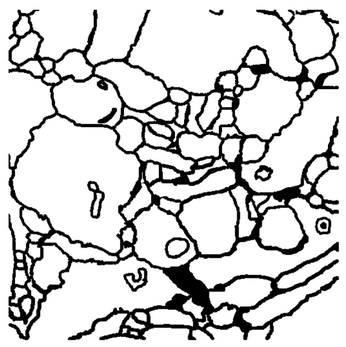
Introduction

The objective of this project is to build a U-Net model that can segment membranes in *electron microscopy* (EM) images. The project utilizes the ISBI Challenge dataset [1], which focuses on segmentation of neuronal structures in EM stacks. Afterwards the pipeline will be used to detect myelinated axons of human nerves.

Input image



Masked image



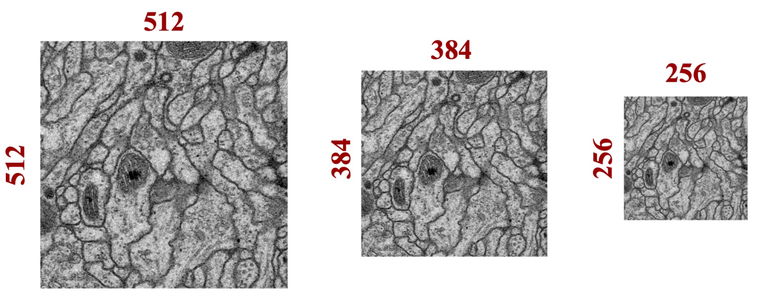
Data Preprocessing

Split into train/validation/test

To enable hyperparameter tuning, the original training set is further split into separate training and validation subsets.

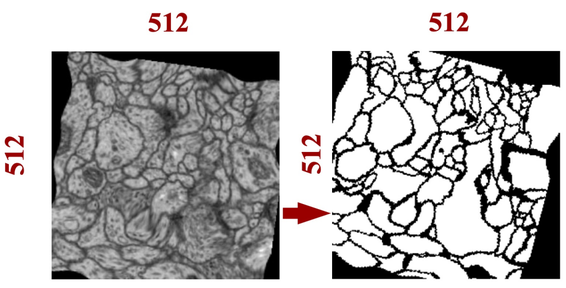
Resize 512/256/128

To investigate the effect of image size on the U-Net, all images will either be resized or kept at their original size.



Data augmentation

To expand the dataset and improve the robustness of the U-Net model, data augmentation techniques are applied. See the Model section for parameters.



Augmentation images

From the 25 train images a total of 250 augmented images are produces.

Model

Model Parameters

Epochs: 100
Learning Rate: 0.0005
Loss Function: Focal Loss
Optimizer: Adam
Batch Size: 16
Patience: 10
Min Delta: 0.001

U-Net Settings

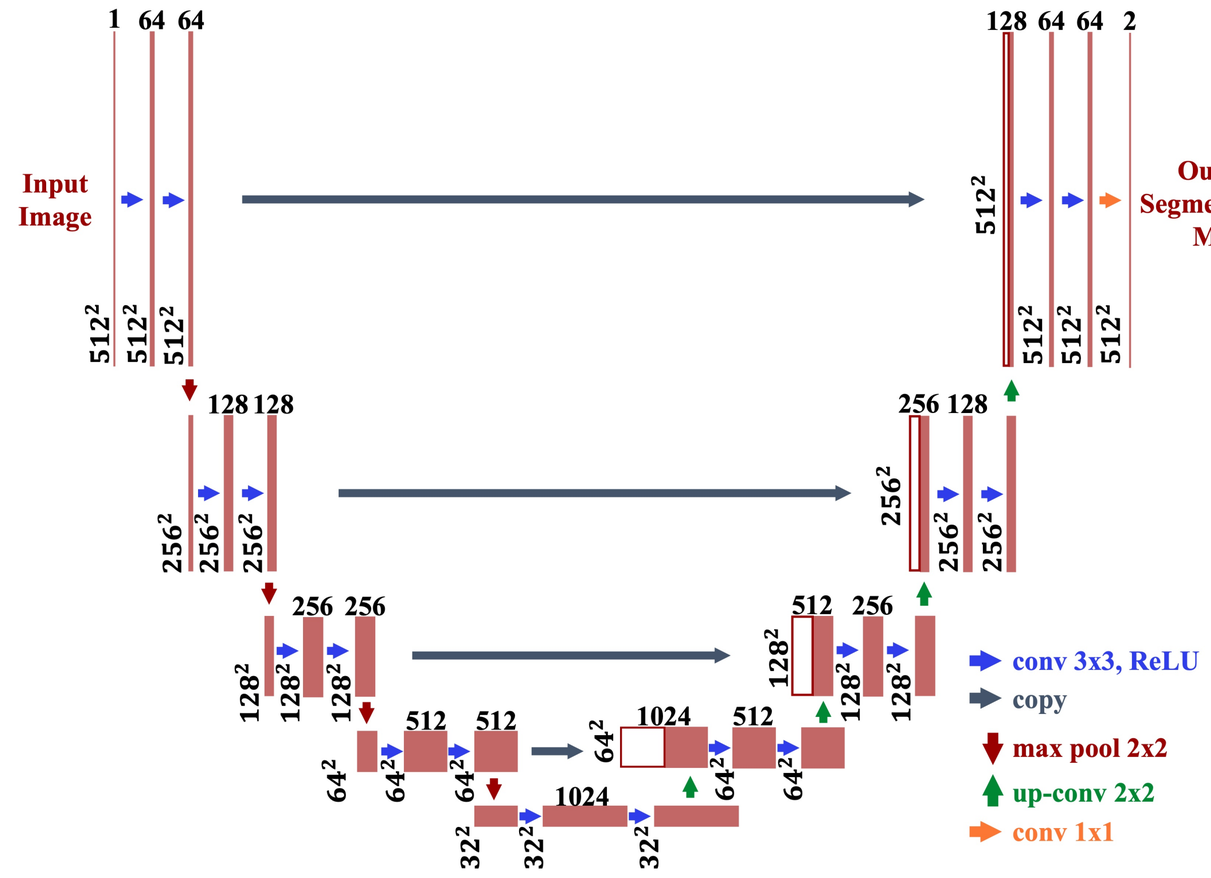
Padding: Same
Size with Patches: 256, 384
Size without Patches: 256, 384, 512

Augmentation

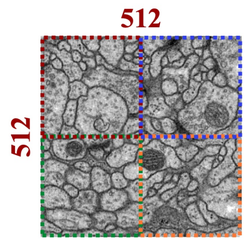
Elastic Deformation (with prob 25%)
Alpha: 3.0
Sigma: 15.0

Gray Variation (with prob 25%)
Brightness range: 0.1
Contrast Range: 0.1

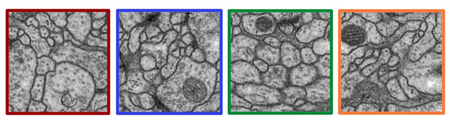
Random Rotation: [-15°, 15°]
Random Crop
Normalization



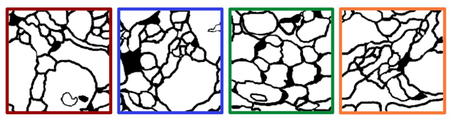
Patches



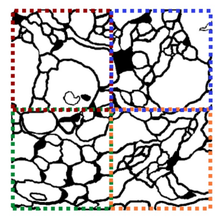
Split the input image into smaller patches.



Send each patch through the model to get a prediction.

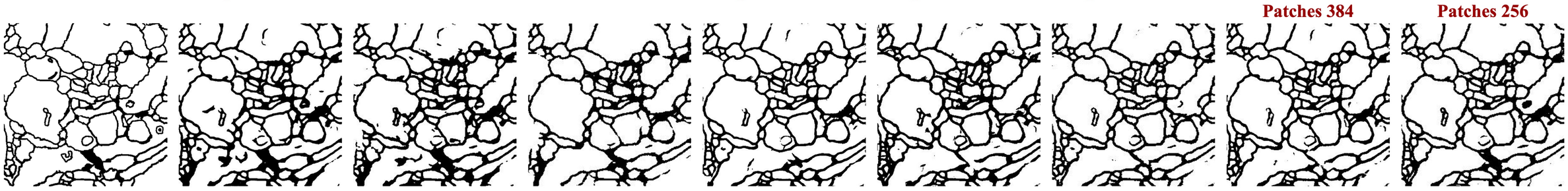


Combine all patch predictions into the final prediction.



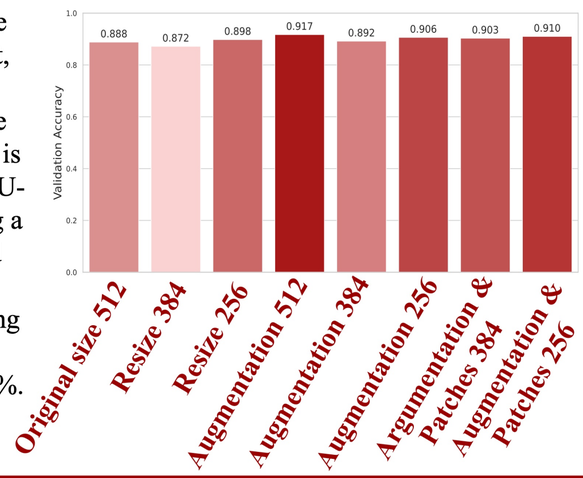
Results

Below are the predicted masks for the validation set from all developed models, with the corresponding ground truth masks shown on the left. The first three models were trained on resized and original image sizes without augmentation. The next three models used data augmentation during training, and the final two models were also trained with augmentation, but inference was performed using image patches.



Accuracy of the models

As shown in the plot on the right, the validation accuracy for the different models is similar, with the U-Net model (using a size of 512 and augmentation applied) achieving the highest accuracy of 91.7%.



Model performance for U-Net 512 with augmentation

Metrics	Value
Accuracy	91.7%
Sensitivity	92.7%
Specificity	87.8%
Dice	94.7%
IoU	90.0%

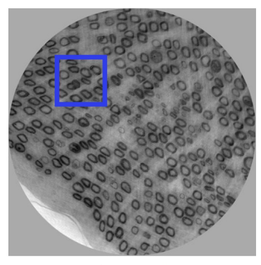
Model performance for U-Net 512 with augmentation

Metrics	Value
Accuracy	96,9%
Sensitivity	76,8%
Specificity	99.7%
Dice	85.7%
IoU	75.4%

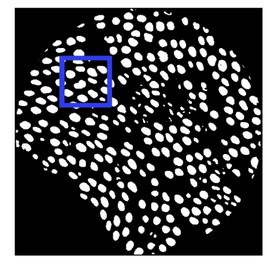
Accuracy measures overall correctness, sensitivity indicates how well it detects positives, and specificity shows its ability to avoid false positives. The Dice coefficient and IoU assess segmentation quality, with higher values indicating better overlap between predicted and actual regions.

The image on the right shows the input image with the masked output for a nerve slice, where the region of interest is marked with a blue box. Below, the myelinated axons entering and exiting this box are visualized.

Input image of a slice of the nerve



Predicted mask for a slice of the nerve





Conclusions

The best-performing model was trained on images at the original size of 512×512. This is likely because reducing the image size introduced pixelation and loss of details, which negatively affected performance. Data augmentation consistently improved accuracy across all models. Using patches during inference did not have a significant impact, possibly due to the relatively small image size of 512×512. Furthermore, it can be concluded that the same pipeline used for the EM images can be applied to segment the nerve dataset, yielding a good performance.

References

1. Arganda-Carreras et al. (2015). *Crowdsourcing image segmentation algorithms for connectomics. Front. Neuroanat. 9:142*

2. Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation.* In *MICCAI 2015, LNCS vol. 9351*